as aquatic respiratory organ (**cutaneous respiration**). Dissolved oxygen in the water gets, exchanged through the skin by diffusion. On land, the buccal cavity, skin and lungs act as the respiratory organs. In **buccal respiration** on land, the mouth remains permanently closed while the nostrils remain open. The floor of the buccal cavity is alternately raised and lowered, so air is drawn into and expelled out of the buccal cavity repeatedly through the open nostrils. Respiration by lungs is called **pulmonary respiration**. The lungs are a pair of elongated, pink coloured sac-like structures present in the upper part of the trunk region (thorax). Air enters through the nostrils into the buccal cavity and then to the lungs. During **aestivation** and **hibernation** gaseous exchange takes place through skin.

The Blood-Vascular System

Blood vascular system consists of a **heart** with three chambers, **blood vessels** and **blood**. Heart is covered by a double-walled membrane called **pericardium**. There are two thin walled anterior chambers called auricles (Atria) and a single thick walled posterior chamber called ventricle. **Sinus venosus** is a large, thin walled, triangular chamber, which is present on the **dorsal side** of the heart. **Truncus arteriosus** is a thick walled and cylindrical structure which is obliquely placed on the **ventral surface** of the heart. It arises from the ventricle and divides into right and left **aortic trunk,** which is further divided into **three aortic arches** namely carotid, systemic and pulmo-cutaneous. The **Carotid** trunk supplies blood to the anterior region of the body. The **Systemic trunk** of each side is joined posteriorly to form the **dorsal aorta.** They supply blood to the posterior part of the body. **Pulmo-cutaneous trunk** supplies blood to the lungs and skin. Sinus venosus receives the deoxygenated blood from the body parts by two anterior precaval veins and one post caval vein. It delivers the blood to the right auricle; at the same time left auricle receives oxygenated blood through the pulmonary vein. Renal portal and hepatic portal systems are seen in frog (Figure. 4. 19 and 4. 20).

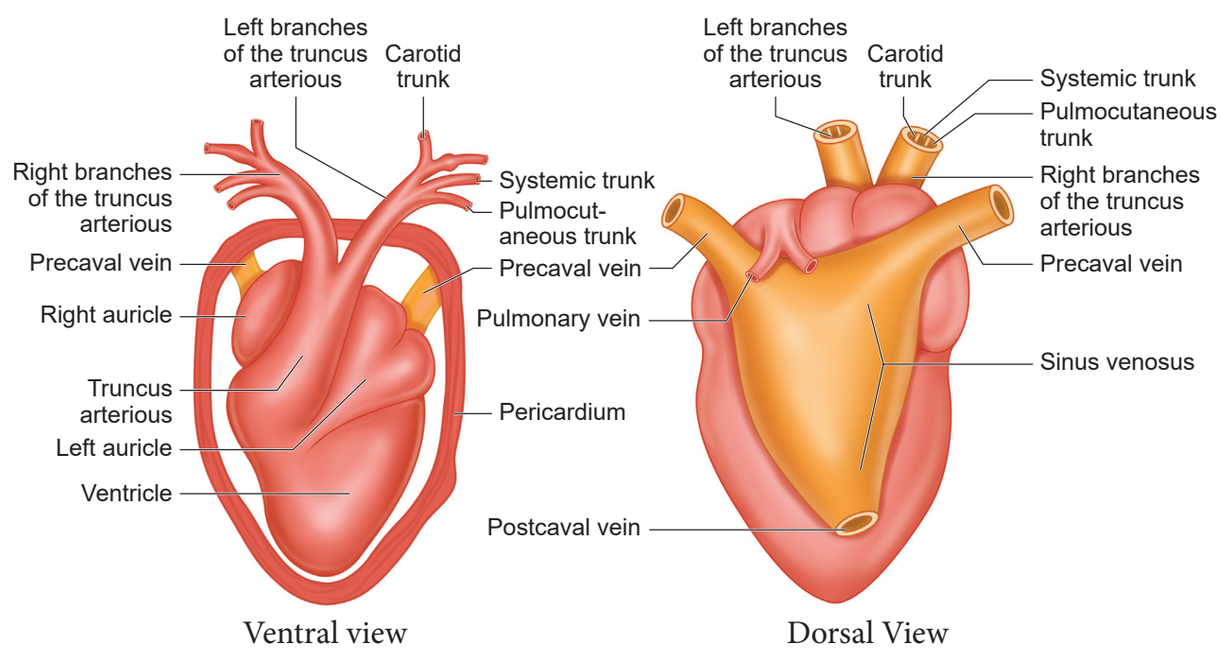


Figure 4.19 *Rana hexadactyla* - Structure of Heart

TN_GOVT_XI_Zoology_Ch04.indd 66 09-01-2020 5.14.27 PM